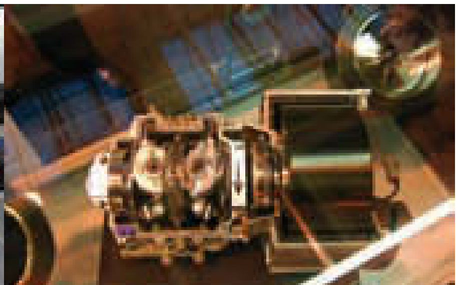
Pit stops are a huge part of the strategy used by teams to gain advantages over the competition, and can be the difference between coming first or fourteenth even...



Just some of the excitement at the average F1 race that we gear heads live for...

The brakes

The need to slow down

Slowing down, or stopping an F1 car is serious stuff. Like road cars, F1 cars also use disc brakes. This is how they work – a rotating disc is flanked by two brake pads, one on either side. When brakes are applied, these pads converge onto the rotating disc, forcing it to slow down. Plenty of heat is generated, and sometimes you may have even seen the brakes glow red-hot. While brakes' sizes are regulated, performance depends on how the individual driver uses them. Particularly when it comes to how the brakes will respond in the latter part of the race.

The fuel

Correct mixture

Long gone are the days when benzene, alcohol and aviation fuel were mixed to run an F1 car. In an attempt to make it safer, the fuel now used is very close to what we see in a road car. However, with refueling now banned during races, the close link with engine's capability to drink less fuel gains paramount importance. An F1 driver can change the fuel mixture during the race, to either enhance engine performance or to consume less fuel.

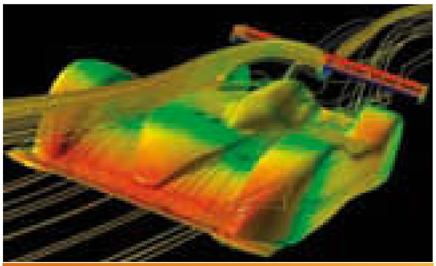
The tyres are supposed to be as bald as this, and melt while driving

Tyres are kept warm, so that they can just be put on and raced with

KERS

Kinetic Energy Recovery System

Simple thinking here – capture the energy released during braking, and use it to offer an acceleration boost with the battery power. FIA regulations state that you can use KERS for about five seconds per lap, offering 80 bhp extra power during that time. The driver has the control for this. However, the size of the KERS unit is up to what the individual teams decide. Surprisingly, Red Bull Racing had the smallest KERS unit this season.



The entire design philosophy of an F1 car is to reduce drag and create downforce